

Security+ 3.2 · Secure Communication

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.2 — Secure Communication (9:55)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: VPN: encrypted tunnel over a public network — IPsec at Layer 3 (network); TLS/SSL at Layer 7 (application). Site-to-site VPN connects two offices. SD-WAN: software-defined — routes by application. SASE: converges everything to the cloud (VPN + NGFW + proxy + CASB + ZTNA).

High-Yield Core Concepts Tested on Exam

- IPsec vs TLS: Layer 3 vs application layer
- SD-WAN: dynamic application-aware routing and quality
- SASE: networking + security as a cloud service (the emerging model)

Exam Tips & Classic Scenarios

- 'encrypted tunnel over public network' = VPN

Typical Exam Scenario: 'connects two offices securely over the internet' = site-to-site VPN (IPsec).

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.2 — Secure Communication)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **IPSEC:** Internet Protocol Security: Network layer (L3) security suite consisting of Authentication Header (AH for integrity), Encapsulating Security Payload (ESP for encryption), and IKE for key exchange.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Protocol that creates an encrypted tunnel to the corporate network over the internet:

- A)** VPN (e.g., IPsec/TLS)
- B)** DNS
- C)** DHCP
- D)** SNMP

2. Accelerates content delivery by serving copies on servers near the user:

- A)** CDN (Content Delivery Network)
- B)** Load balancer
- C)** Firewall
- D)** Bastion

3. TLS/SSL are protocols used to protect data:

- A)** At-rest
- B)** In-transit
- C)** In use
- D)** In backup

4. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'IPsec vs TLS: layer 3 vs application'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: IPsec vs TLS: layer 3 vs application...

5. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'SD-WAN: dynamic per-application routing and quality'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: SD-WAN: dynamic per-application routing and quality...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'SASE: network + security as a cloud service (the new model)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: SASE: network + security as a cloud service (the new model)...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'encrypted tunnel over public network'=VPN'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'encrypted tunnel over public network'=VPN...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 3.2) exam, an analyst evaluates the following technical requirement: 'connects two offices securely over the internet'=site-to-site VPN (IPsec)'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'connects two offices securely over the internet'=site-to-site VPN (IPsec)....
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'IPsec vs TLS: layer 3 vs application'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: IPsec vs TLS: layer 3 vs application...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Secure Communication): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.2), an analyst evaluates the following technical requirement: 'SD-WAN: dynamic routing by application and quality'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: SD-WAN: dynamic routing by application and quality...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — VPN (e.g., IPsec/TLS).

✓ **Why it's correct:** A VPN (Virtual Private Network) creates an encrypted tunnel over a public network (such as the internet) to allow remote users to securely access the corporate network. Protocols such as IPsec (network layer) or TLS (transport layer) are used to establish this tunnel, ensuring confidentiality, integrity, and authentication. CompTIA Security+ defines VPN as the standard technology for secure remote access, essential for remote workers and branch offices. The option 'VPN (e.g., IPsec/TLS)' is the only one that correctly describes this mechanism.

✗ **Why the others are wrong:**

- *DNS:* DNS (Domain Name System) is a protocol for resolving domain names to IP addresses. It does not create encrypted tunnels or provide secure access to corporate networks. Although security extensions like DNSSEC exist, they aim to authenticate DNS responses, not establish private connections. DNS is an infrastructure service, not a tunneling mechanism. Therefore, it does not meet the requirement of creating an encrypted tunnel to the corporate network.
- *DHCP:* DHCP (Dynamic Host Configuration Protocol) is used to automatically assign IP addresses and other network configurations to devices. It does not provide encryption or tunneling. It is a network management protocol, essential for configuration, but not for secure remote access. There is no relation to creating encrypted tunnels. Therefore, it is incorrect.
- *SNMP:* SNMP (Simple Network Management Protocol) is used for monitoring and managing network devices, such as routers and switches. It allows collecting metrics and configuring equipment, but does not establish encrypted tunnels for user access. Although SNMPv3 offers security (authentication and encryption), its purpose is management, not remote access to the corporate network. Therefore, it is not the correct answer.

2) A — CDN (Content Delivery Network).

✓ **Why it is correct:** A CDN (Content Delivery Network) is a global network of edge servers that cache copies of static and dynamic content, delivering them from the point geographically closest to the user. This reduces latency, improves load times, and decreases load on the origin server. In CompTIA Security+ (domain 3.4), CDN is the standard solution for accelerating content delivery, exactly as described in the prompt: 'accelerates content delivery by serving copies on servers near the user'.

✗ **Why the others are wrong:**

- *Load balancer:* A load balancer distributes traffic across multiple servers in the same region or data center, but does not replicate content on geographically distributed servers near the user. Its function is load balancing and high availability, not acceleration by geographic proximity. Therefore, it is not the correct answer.
- *Firewall:* A firewall is a security device that filters network traffic based on rules, blocking unauthorized access. It does not store content copies or accelerate delivery; its function is access control and protection. Thus, it is incorrect for the context.
- *Bastion:* A bastion (or jump host) is a secure server used as an intermediary access point to manage resources in private networks, minimizing exposure. It does not serve content or accelerate delivery; it is an access security measure. Therefore, it does not meet the prompt.

3) B — In-transit.

✓ **Why it is correct:** TLS (Transport Layer Security) and its predecessor SSL (Secure Sockets Layer) are cryptographic protocols that operate at the transport and application layers of the OSI/TCP-IP model. They are designed to protect data in-transit — that is, during transmission across the

network. TLS/SSL provide confidentiality (symmetric encryption), integrity (HMAC), and authentication (digital certificates) for communications such as HTTPS, FTPS, and SMTPS. In CompTIA Security+, TLS/SSL are classic examples of controls that protect data in motion, mitigating attacks such as eavesdropping, man-in-the-middle, and tampering.

✗ Why the others are wrong:

- *At-rest*: At-rest encryption protects data stored on disk, in databases, or in backups. TLS/SSL do not act on data at rest, because they are communication protocols that only engage during transmission. This option is wrong because it confuses the data state: TLS/SSL do not encrypt files sitting idle, but rather the data flow between systems.
- *In use*: In-use encryption protects data while it is being processed in memory, using techniques such as secure enclaves (Intel SGX) or homomorphic encryption. TLS/SSL do not operate in this state, because they have no access to process memory, only to the communication channel. This option is incorrect because it describes a different data state than the one protected by TLS/SSL.
- *In backup*: Backups are copies of data stored for recovery. Although they can be protected by at-rest encryption, TLS/SSL are not used to encrypt backups themselves, but rather to protect the transmission of backup data across the network. This option is wrong because it restricts the scope to backups, whereas TLS/SSL protect any data in transit, regardless of whether it is a backup or not.

4) D — Implement the corresponding solution: IPsec vs TLS: layer 3 vs application....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): IPsec vs TLS: layer 3 vs application.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: SD-WAN: dynamic per-application routing and quality....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): SD-WAN: dynamic per-application routing and quality.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: SASE: network + security as a cloud service (the new model)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): SASE: network + security as a cloud service (the new model).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: 'encrypted tunnel over public network'=VPN....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): 'encrypted tunnel over public network'=VPN. A VPN (Virtual Private Network) creates a secure, encrypted tunnel over an untrusted network such as the internet, ensuring confidentiality and integrity of data in transit. The correct answer directly reflects this concept by recommending the implementation of the corresponding solution.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: 'connects two offices securely over the internet'=site-to-site VPN (IPsec).....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): 'connects two offices securely over the internet'=site-to-site VPN (IPsec). A site-to-site VPN using IPsec creates an encrypted tunnel between two network locations over an untrusted network like the internet, ensuring confidentiality and integrity of data in transit.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) A — Implement the corresponding solution: IPsec vs TLS: layer 3 vs application....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): IPsec vs TLS: layer 3 vs application. IPsec operates at the network layer (Layer 3) to secure IP communications, while TLS operates at the application layer (Layer 7) to secure application-level protocols such as HTTPS, providing encryption and integrity for data in transit.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) D — Implement the corresponding solution: SD-WAN: dynamic routing by application and quality....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.2 (Secure Communication): SD-WAN: dynamic routing by application and quality. SD-WAN (Software-Defined Wide Area Network) uses software-defined networking principles to intelligently route traffic across multiple WAN connections based on application requirements, network conditions, and quality of service (QoS) policies. This improves performance, reliability, and security by dynamically selecting the best path for each application. Security considerations include encrypting traffic, segmenting networks, and integrating with cloud security services.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.